SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	N.KISHAN KUMAR	**Reg. No.:**	192424404
Section / Batch:	B.Tech AI &DS	**Date:**	30-07-2026

Code of Conduct

I _____N.kishan kumar______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

1. Morphological Processing for Search Engine

Input Words

- Analyzing
- Analysis
- Analytical

Step 1: Morphological Breakdown

Word	Root	Affix
Analyzing	Analyze	-ing
Analysis	Analyze	-sis
Analytical	Analyze	-ical

Step 2: Classification

- **Analyzing** → Inflectional (Present participle)
- **Analysis** → Derivational (Forms a noun)
- **Analytical** → Derivational (Forms an adjective)

Step 3: Normalization

All words are mapped to the common base form:

Analyze

Final Output

- Analyzing = Analyze + ing
- Analysis = Analyze + sis
- Analytical = Analyze + ical

Normalized Root = Analyze

2. Morphological Parsing for Sentiment Analysis

Input Words

- Disagree
- Agreement
- Agreeable

Step 1: Morphological Breakdown

Word	Prefix	Root	Suffix
Disagree	dis-	agree	—
Agreement	—	agree	-ment
Agreeable	—	agree	-able

Step 2: Classification

- **Disagree** → Derivational
- **Agreement** → Derivational
- **Agreeable** → Derivational

Meaning

- **dis-** gives opposite meaning.
- **-ment** converts verb into noun.
- **-able** converts verb into adjective.

Normalized Root

Agree

3. Morphology-Based Normalization

Input Words

- Govern
- Government
- Governance

Step 1: Morphological Breakdown

Word	Root	Suffix
Govern	Govern	—
Government	Govern	-ment
Governance	Govern	-ance

Step 2: Classification

- **Govern** → Base word
- **Government** → Derivational
- **Governance** → Derivational

Normalization

All words are converted into the common base form:

Govern

Final Output

- Govern = Govern
- Government = Govern + ment
- Governance = Govern + ance

Normalized Root = Govern

4. Morphological Parsing for Document Classification

Input Words

- Activate
- Activation
- Reactivation

Step 1: Morphological Breakdown

Word	Prefix	Root	Suffix
Activate	—	Act	-ivate
Activation	—	Act	-ivation
Reactivation	re-	Act	-ivation

Step 2: Classification

- **Activate** → Derivational
- **Activation** → Derivational
- **Reactivation** → Derivational

Meaning

- **re-** means "again".
- **-ation** changes verb into noun.

Normalization

All words are mapped to:

Act

Final Output

- Activate = Act + ivate
- Activation = Act + ivation
- Reactivation = Re + Act + ivation

Normalized Root = Act

5. Morphology-Based Normalization

Input Words

- Create
- Creates
- Creating

Step 1: Morphological Breakdown

Word	Root	Suffix
Create	Create	—
Creates	Create	-s
Creating	Create	-ing

Step 2: Classification

- **Create** → Base word
- **Creates** → Inflectional (Present tense)
- **Creating** → Inflectional (Present participle)

Normalization

All words are mapped to:

Create

Final Output

- Create = Create
- Creates = Create + s
- Creating = Create + ing

Normalized Root = Create

Conclusion

Morphological analysis helps identify roots, prefixes, and suffixes while distinguishing between inflectional and derivational forms.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________